

A

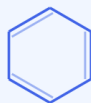
ChEMBL36

The open chemical database

2,400,000
molecules

Encoded as SELFIES strings:

[C] [C] [Branch1] [C] [=O] ...

**B**

APE Tokenisation

Atom Pair Encoding

Input:

[C] [C] [Branch1] [C] [=O] ...

APE merges

Output:

[C][C] [Branch1][C] [=O] ...

C

ModernBERT

RoPE · GeGLU · Local/Global Attention



RoPE positional encoding

alternating local/global attention

GeGLU feed-forward blocks

pre-normalized transformer layers

MLM pretraining on ~2.4M ChEMBL36 SELFIES
APE vocabulary → frozen embeddings

D

Relative Benchmark Performance

MoleculeNet & TDC classification tasks



MoleculeNet
(Molecular property
benchmarks)



7 classification datasets



Biophysical &
physicochemical endpoints



ROC-AUC



TDC
(Therapeutics Data
Commons)



18 classification datasets



Toxicity, ADME, HTS



ROC-AUC